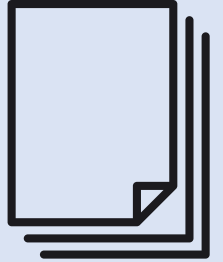


Starter



Find the general solution to:

1.
$$\frac{dy}{dx} + 2y = x$$

2.
$$\ddot{x} + \dot{x} - 6x = 0$$

3. Find the particular solution to

$$\frac{d^2z}{dt^2} - 9z = 0, \text{ given } \left. \frac{dz}{dt} \right|_{t=0} = 16, z|_{t=0} = 4$$

Solutions

1. IF = e^{2x} , $\therefore e^{2x} \frac{dy}{dx} + 2e^{2x}y = xe^{2x}$

$$e^{2x}y = \int xe^{2x} dx = \frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x} dx$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c$$

$$\therefore y = \frac{1}{2}x - \frac{1}{4} + ce^{-2x}$$

Solutions

$$2. \text{ AE} = m^2 + m - 6 = 0, m \in \{2, -3\},$$

$$\therefore x = Ae^{2t} + Be^{-3t}$$

Solutions

$$3. \quad AE = m^2 - 9 = 0, \quad m = 3, \quad \therefore x = Ae^{3t} + Bte^{3t}$$

$$\dot{x} = 3Ae^{3t} + Be^{3t} + 3Bte^{3t}$$

$$x|_{t=0} = 4 = A + 0 \Rightarrow A = 4$$

$$\dot{x}|_{t=0} = 12 = 3A + B + 0 \Rightarrow B = 4$$

$$\therefore x = 4e^{3t} + 4te^{3t}$$

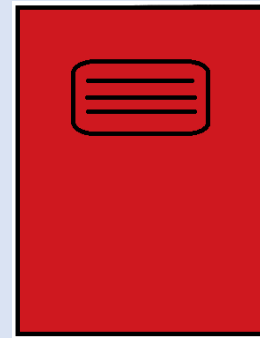
Differential Equations 2

2nd Order Homogeneous 2

Purely Imaginary Roots

Find the general solution to

$$\frac{d^2y}{dx^2} + 9y = 0$$

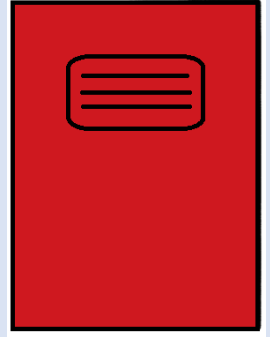


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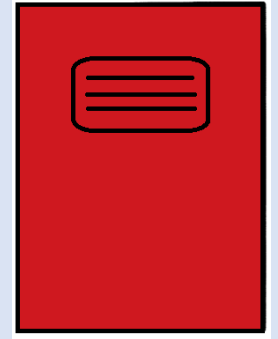
$$y = Ae^{3ix} + Be^{-3ix}$$

$$= A(\cos 3x + i \sin 3x) + B(\cos -3x + i \sin -3x)$$

$$= A(\cos 3x + i \sin 3x) + B(\cos 3x - i \sin 3x)$$

$$= (A + B) \cos 3x + (A - B) i \sin 3x$$

$$= C \cos 3x + D \sin 3x$$



Purely Imaginary roots in General

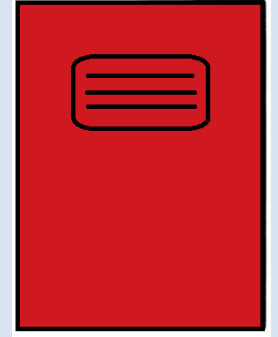
Show that the general solution to

$$\frac{d^2y}{dx^2} + \omega^2 y = 0$$

can be written in the form

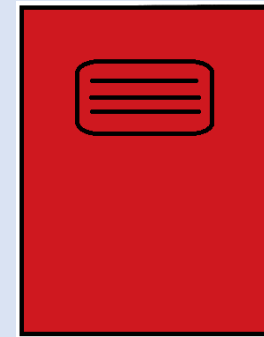
$$y = A \cos \omega x + B \sin \omega x$$

where A, B, ω are real numbers.



Complex roots

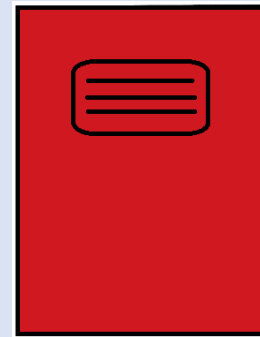
Can we extend this to a complex root, $\lambda \pm \mu i$?



Complex roots

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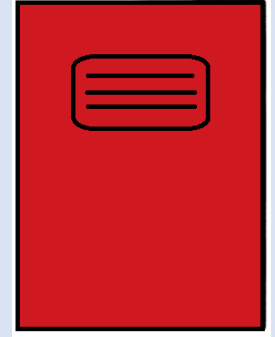
$$\begin{aligned} y &= Ae^{(\lambda + \mu i)x} + Be^{(\lambda - \mu i)x} \\ &= Ae^{\lambda x} e^{\mu i x} + Be^{\lambda x} e^{-\mu i x} \\ &= e^{\lambda x} (Ae^{\mu i x} + Be^{-\mu i x}) \\ &= e^{\lambda x} (C \cos \mu x + D \sin \mu x) \end{aligned}$$



Sketching the solutions

It's natural that when we look at either of the two solution forms, we want to know what they look like!

Harmonic forms from A Level Maths can give us an answer.

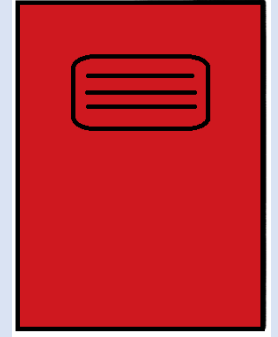
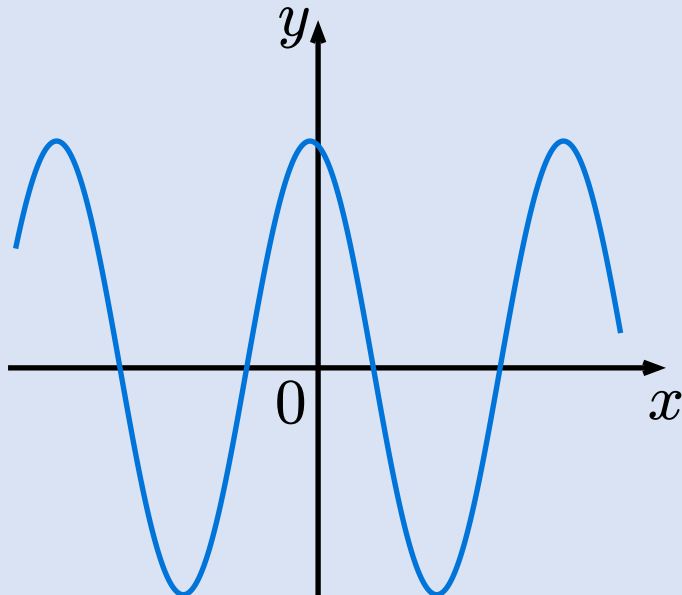


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$$y = A \cos \mu x + B \sin \mu x \equiv R \cos(x + \alpha)$$

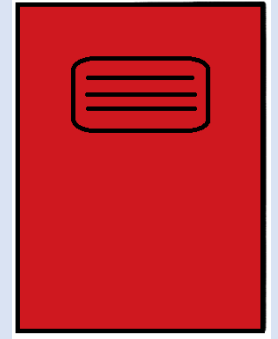
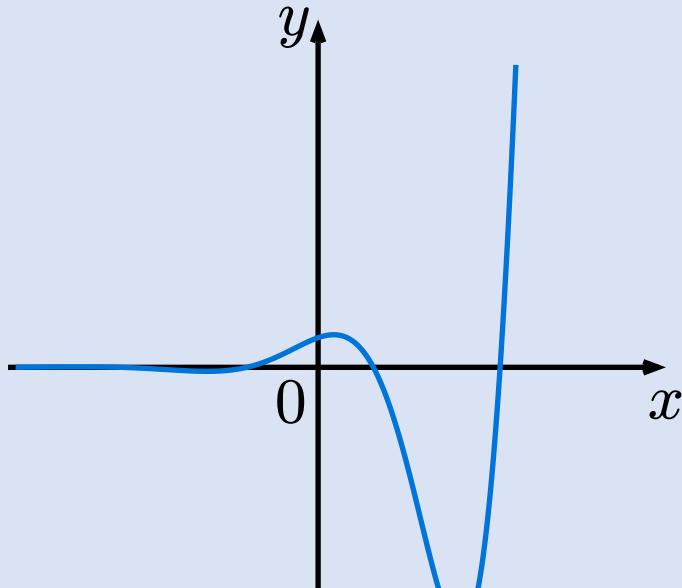


Sketching the solutions

It's natural that when we look at either of the two solution forms, we want to know what they look like!

Harmonic forms from A Level Maths can give us an answer.

$$y = e^{\lambda x} (A \cos \mu x + B \sin \mu x) \equiv R e^{\lambda x} \cos(x + \alpha)$$

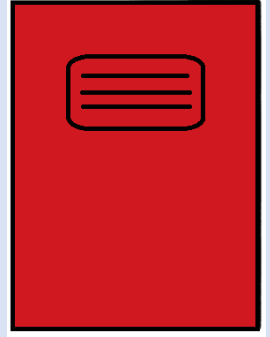


Finding a Particular Solution

Find and sketch the particular solution to

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$$\text{Initially } x = 1, \dot{x} = 2$$



Finding a Particular Solution

Find and sketch the particular solution to

$$\ddot{x} - 2\dot{x} + 2x = 0$$

$$\text{Initially } x = 1, \dot{x} = 2$$

$$\text{AE: } m^2 - 2m + 2 = 0 \Rightarrow m = 1 \pm i$$

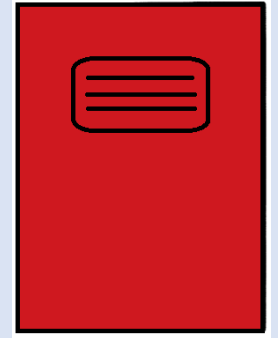
$$\therefore x = e^t (A \cos t + B \sin t)$$

$$\therefore \dot{x} = e^t (A \cos t + B \sin t) + e^t (B \cos t - A \sin t)$$

$$x|_{t=0} = 1 = e^0 (A \cos 0 + B \sin 0) \Rightarrow A = 1$$

$$\dot{x}|_{t=0} = 2 = e^0 (A \cos 0 + B \sin 0) + e^0 (B \cos 0 - A \sin 0) \Rightarrow B = 1$$

$$\therefore x = e^t (\cos t + \sin t) = \sqrt{2}e^t \cos\left(t + \frac{\pi}{4}\right)$$

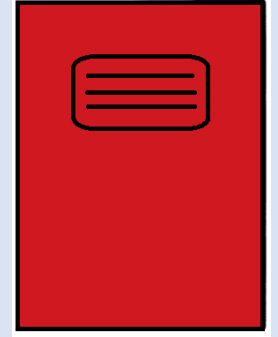
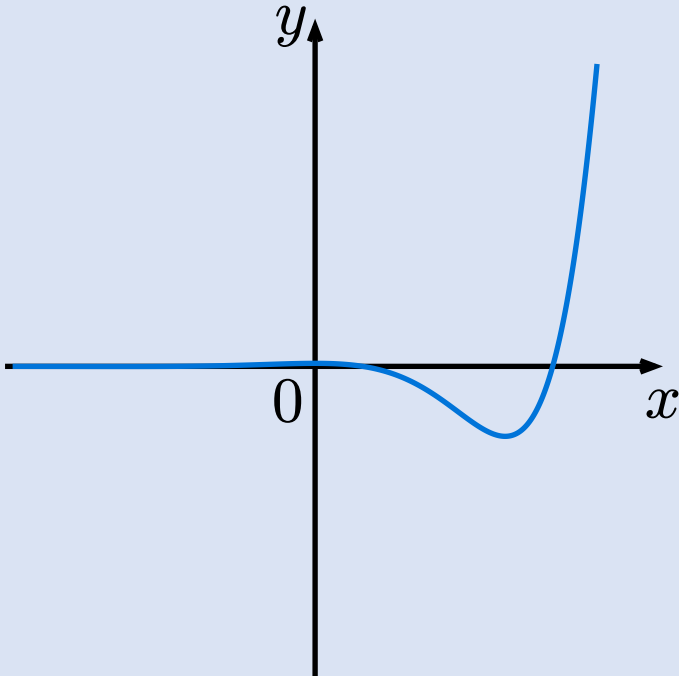


Finding a Particular Solution

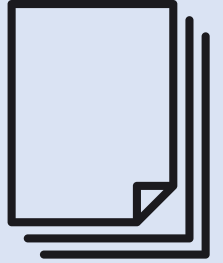
Find and sketch the particular solution to

$$\ddot{x} - 2\dot{x} + 2x = 0$$

Initially $x = 1, \dot{x} = 2$



Recall



Given the equation

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

state the four possible forms the general solution may take, and the conditions under which it will take them.

General Solutions

If the auxiliary equation, $am^2 + bm + c = 0$ have:

- Two distinct real roots, λ, μ :

$$y = Ae^{\lambda x} + Be^{\mu x}$$

- Repeated real roots, λ :

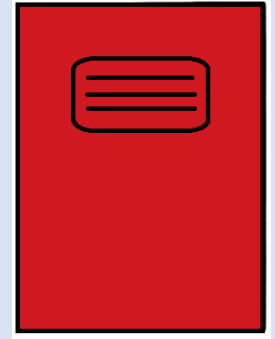
$$y = Ae^{\lambda x} + Bxe^{\lambda x}$$

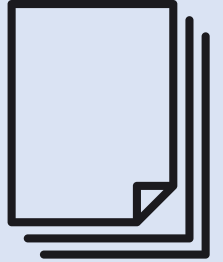
- Purely imaginary roots, $\pm \omega i$:

$$y = A \cos \omega x + B \sin \omega x \text{ or } R \cos(\omega x + \alpha)$$

- Complex roots, $\alpha \pm \beta i$:

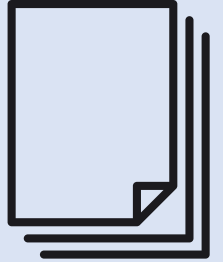
$$y = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$





Practice! (4s.f. where appropriate)

1. Find the general solution to the equation $\frac{d^2x}{dt^2} + 18\frac{dx}{dt} + 81x = 0$.
2. Find the general solution to the equation $\frac{d^2y}{dx^2} + 20\frac{dy}{dx} + 100y = 0$.
3. Find the general solution to the equation $5\ddot{x} + 3x = 0$.
4. Find the general solution to the equation $\ddot{x} + 18\dot{x} + 81x = 0$.
5. Find the general solution to the equation $5\frac{d^2p}{dq^2} + 3\frac{dp}{dq} + 10p = 0$.
6. Find the general solution to the equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 8y = 0$.



Practice! (4s.f. where appropriate)

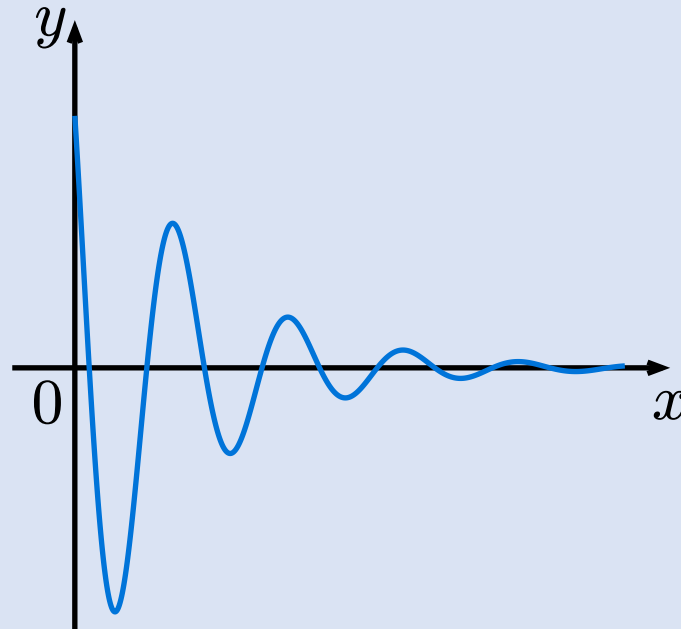
1. $x = Ae^{-9t} + Bte^{-9t}$
2. $y = Ae^{-10x} + Bxe^{-10x}$
3. $x = A \sin(0.7746t) + B \cos(0.7746t)$
4. $x = Ae^{-9t} + Bte^{-9t}$
5. $p = e^{-\frac{3}{10}q} (A \sin(1.382q) + B \cos(1.382q))$
6. $y = Ae^{-2x} + Be^{-4x}$

Damping

These second order DEs often describe real-world events. If you think about a spring, for instance, its acceleration is based on the distance from its resting position, but so is its velocity.

Many such systems attempt to reach an equilibrium position.

EG:



Damping

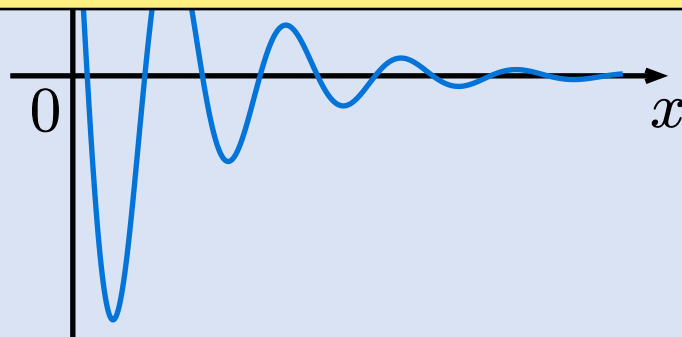
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Moving Examples! 

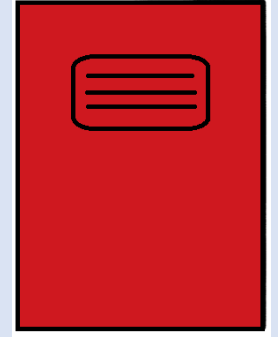


Damping

When discussing these solutions, we consider **under-damped**, **over-damped** and **critically damped** systems.

The DEs all have particular solutions when these occur:

- **under-damped**: auxiliary equation has complex roots.

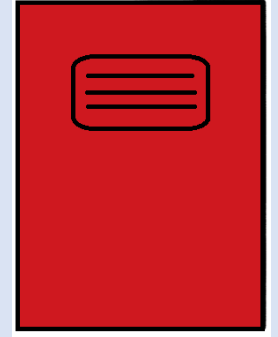


Damping

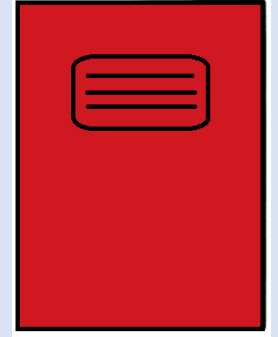
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Damping

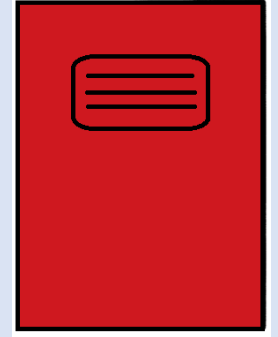


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- **under-damped**: auxiliary equation has complex roots.
- **over-damped**: auxiliary equation has distinct real roots.
- **critically damped**: auxiliary equation has repeated negative roots.
- **undamped** systems are when the AE roots are purely imaginary (Simple Harmonic Motion).